



Features of Images

08

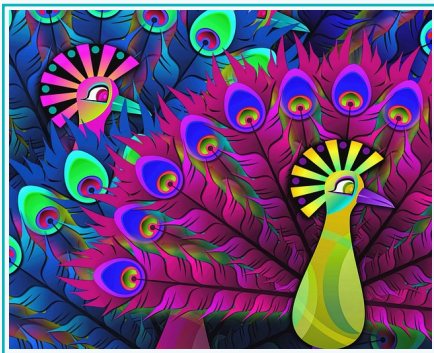
Look Mom, I clicked so many pictures of the Jeddah Port Lighthouse with the digital camera.

These are beautiful pictures. Can I get one to put up as a wallpaper on my phone?

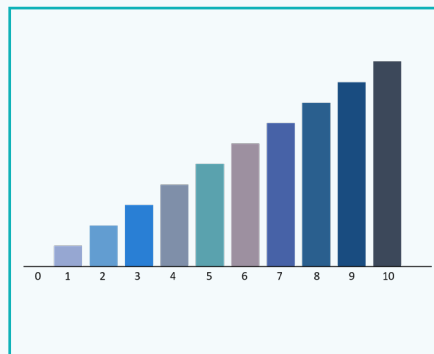


An image is a visual representation of information. It can have many different forms. An image could be a painting, graph, drawing, or a photograph. An image can be used for different purposes. It can be used on a poster, as a wallpaper and for lots of other things.

For example:



Digital Painting



Graph

8. Features of Images

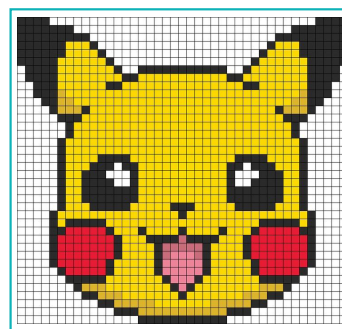
Nowadays, most of the photographs we see are digital. They are taken with a digital camera or smartphone.

A digital Image is made up of **Pixels**.



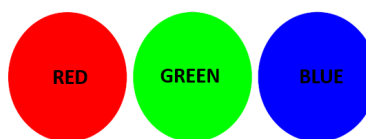
Pixel

A Pixel is a single point in a digital image. On a computer screen, a pixel is usually a square. When we zoom in on an image, we can view its individual pixels. A digital image is made up of thousands of pixels.



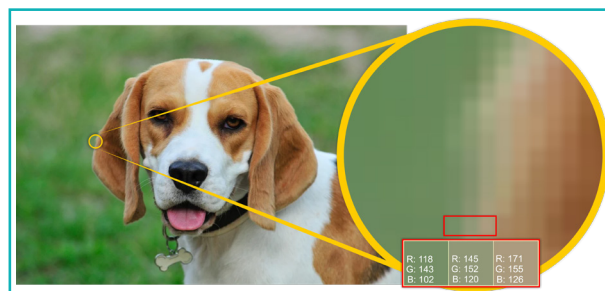
One square = one pixel

Each pixel is a combination of colours. It can be a combination of either RGB (Red, Green, and Blue) or CMYK (Cyan, Magenta, Yellow, and Key colour-black).



An image that is made up of pixels is called a **Raster Image**.

For example:

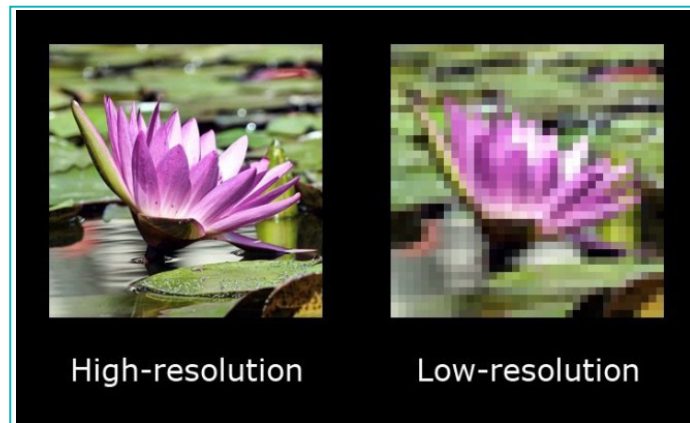


Bytes

RGB is usually used for images that appear on a screen. CMYK is used for images that we print out.

Have you ever wondered why some images appear very clear and some appear patchy & blurred?

This happens because of the number of pixels in an image. The total number of pixels in an image is called its **pixel resolution**. An image with a higher resolution appears clear. However, an image with a low resolution appears blurry when we zoom into it.



Exercise

Fill in the blanks

- 1 A _____ is a single point in a digital picture.
- 2 The total count of pixels in an image is called the _____ of the image.

Activity

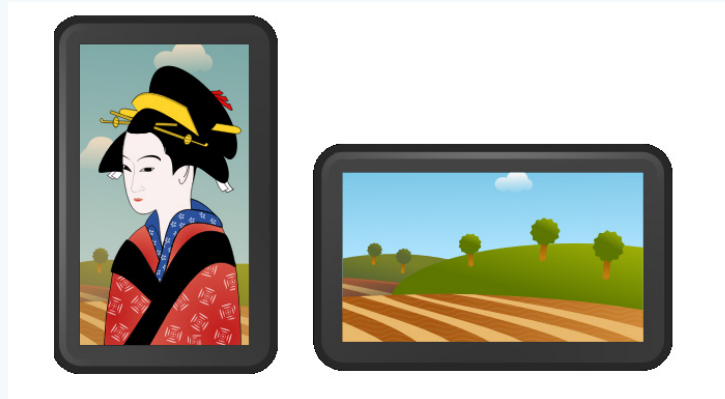
Enhance an image to make it a phone wallpaper

A **landscape image** is a horizontal image. Its width is greater than its height.

A landscape image will not fit correctly as a wallpaper on a mobile phone as it has a vertical screen. We should use a portrait image for a wallpaper for a mobile phone screen.

Activity

A **portrait image** is a vertical image. It's height is more than its width.



Portrait Image

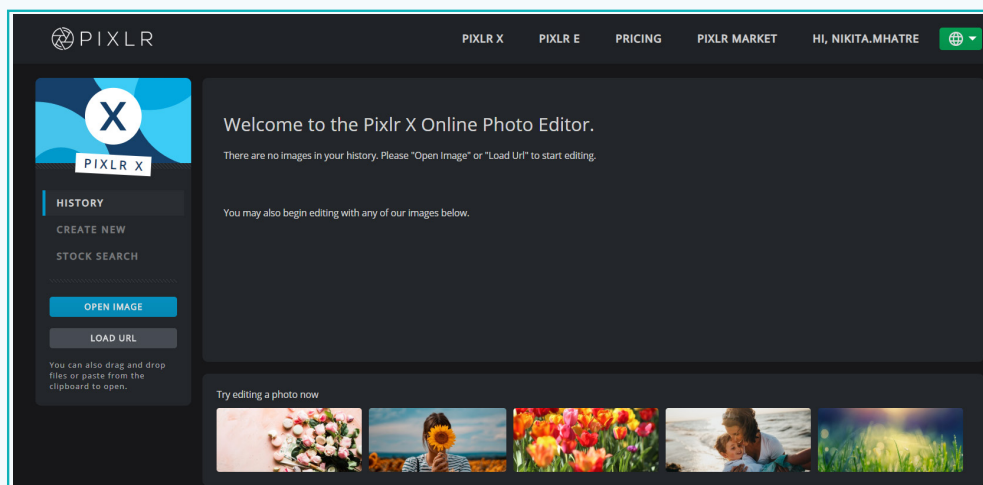
Landscape Image

Image processing is a method to perform some operations on an image to enhance or to extract some useful information from an image. Image processing includes changing various aspects of an image example: height, colour etc.

We can use a computer or a device like a smartphone to enhance an image. There are multiple software and applications we can use for this. For example: Photo editor, GIMP, Photoshop, Pixlr, Snapseed, etc.

For this activity, we will use the Pixlr web application.

Pixlr is a photo editor which we use in a browser. Pixlr is convenient and easy to use. It quickly gives us edited images that look professional.

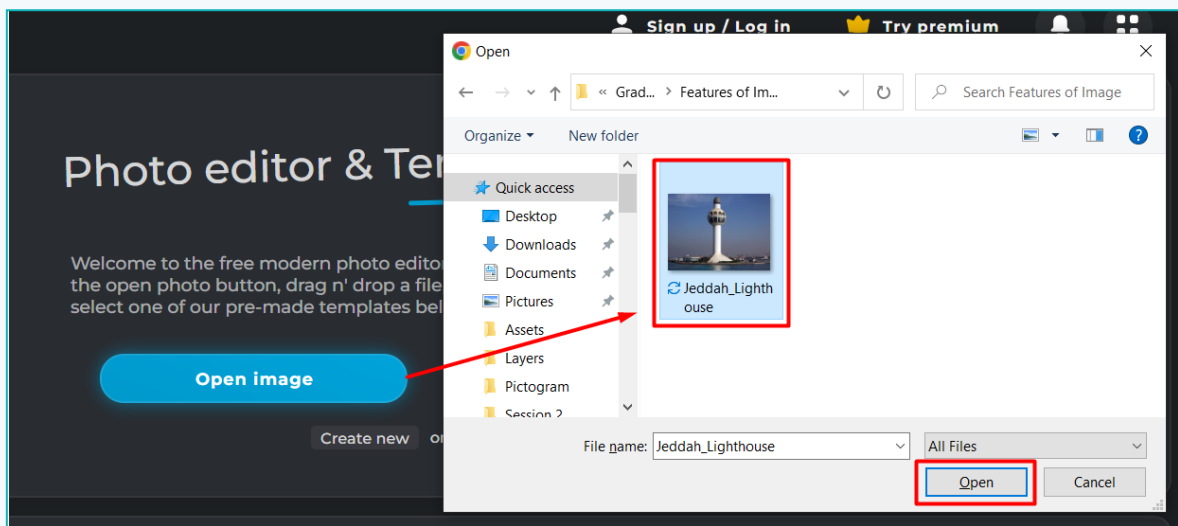


Activity

- 1 Pick out an image to use as a wallpaper.



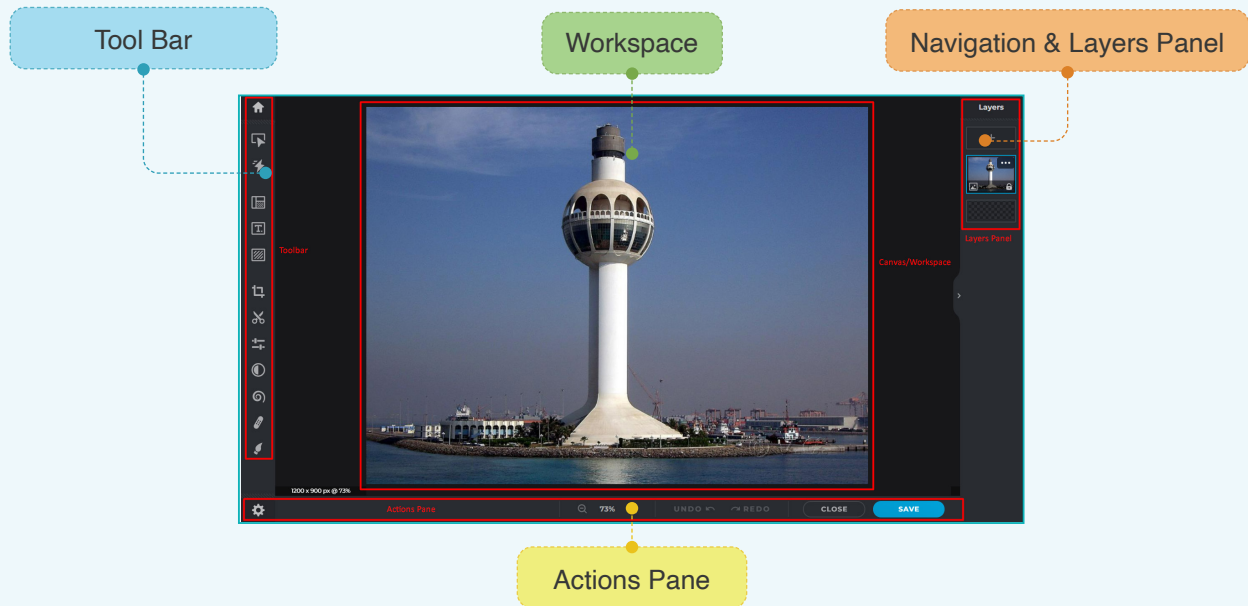
- 2 To load the image in the application, click on Open Image.



- 3 The image will open in 'editor'. This is the editing interface of Pixlr. It is divided into 4 major parts.

Note: We will explore these parts later.

Activity

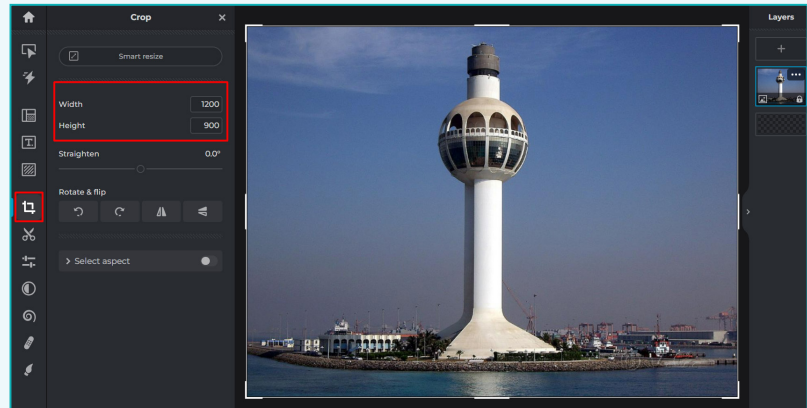


- 4 The image will open in '**Workspace**'. This is where we enhance the image.
 - 5 The image has some unnecessary parts and extra space. We can crop the image to correct this.
- **Cropping** is a process that removes unwanted space on the outside of an image. We can crop our image to turn it into a vertical one.



Activity

- 6 The Crop tool can be found in the tool bar. The Tool Bar consists of all the tools that we can use to edit an image.

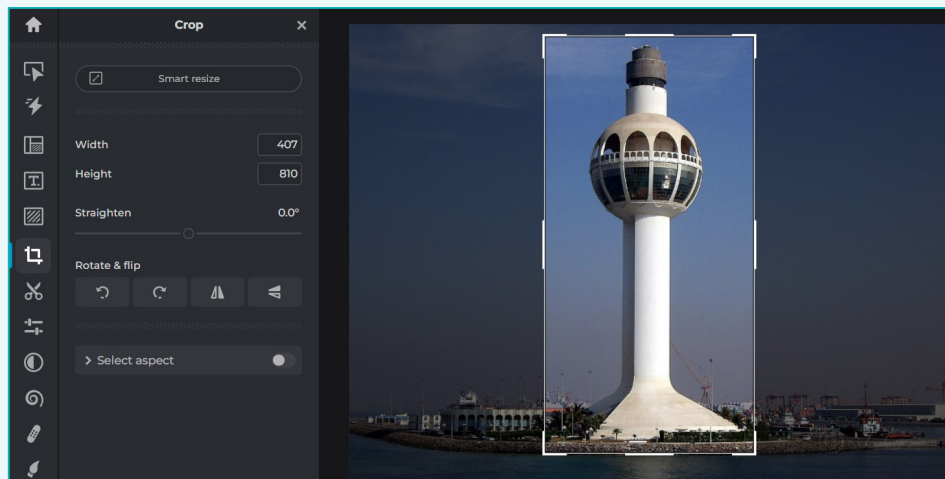


- 7 When the Crop tool is selected, we can view the dimensions of the image. The dimensions consist of the width and height of the image.

Note: The width and height of an image are determined by the number of pixels in it. This image has 1200 pixels along its width (horizontal) and 900 pixels along its height (vertical).

To crop the image

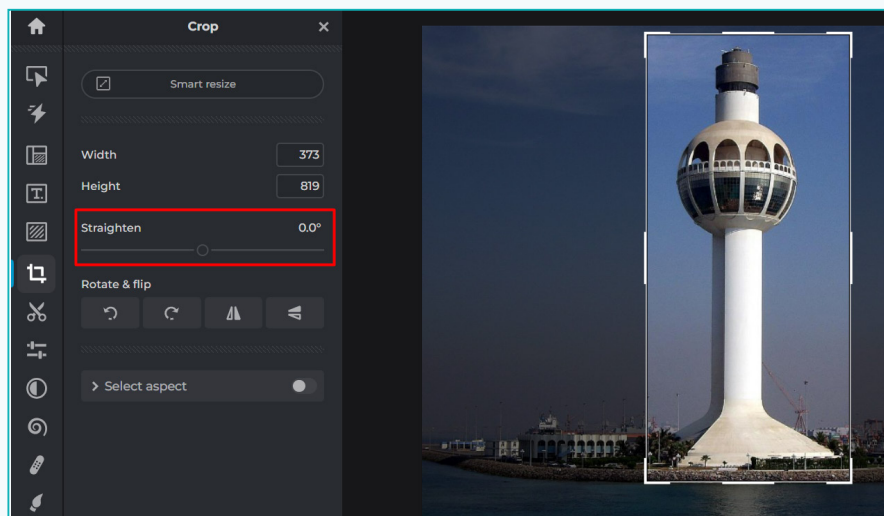
- 8 Drag the sides of the Crop tool to select the required part of the image.



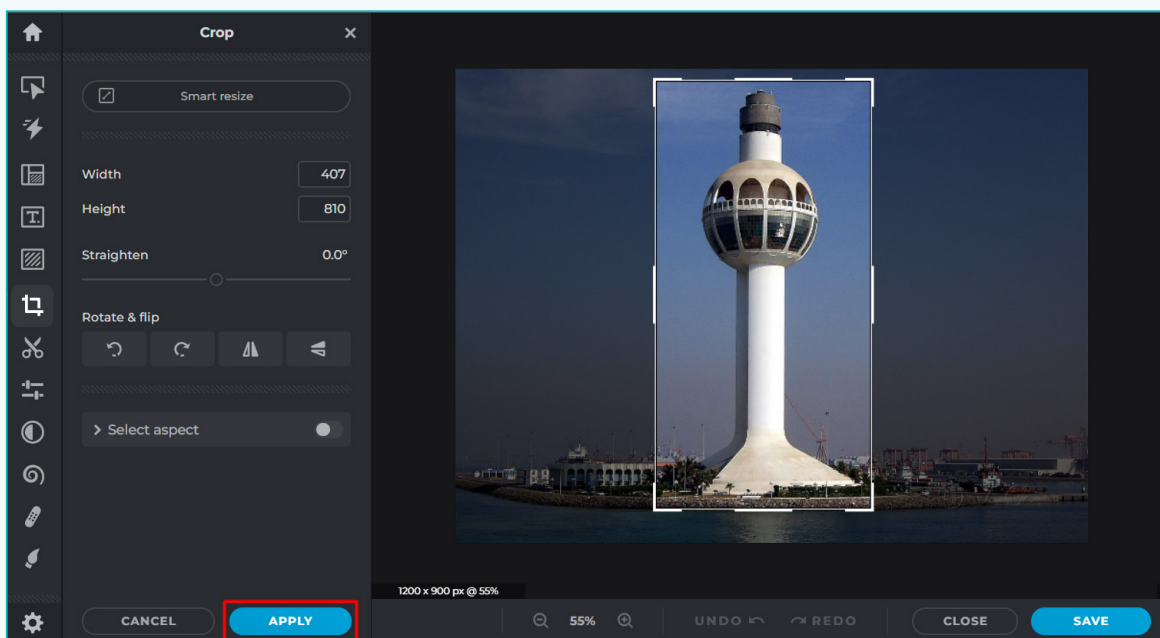
- 9 Observe how the dimensions of the image also change when we crop the image. This happens since we also cut out unwanted pixels from the image. Once the image looks correct, click on 'Apply'.

Activity

- 10 If the image seems slightly tilted, we can straighten the image. This will make it fit the mobile's screen perfectly.



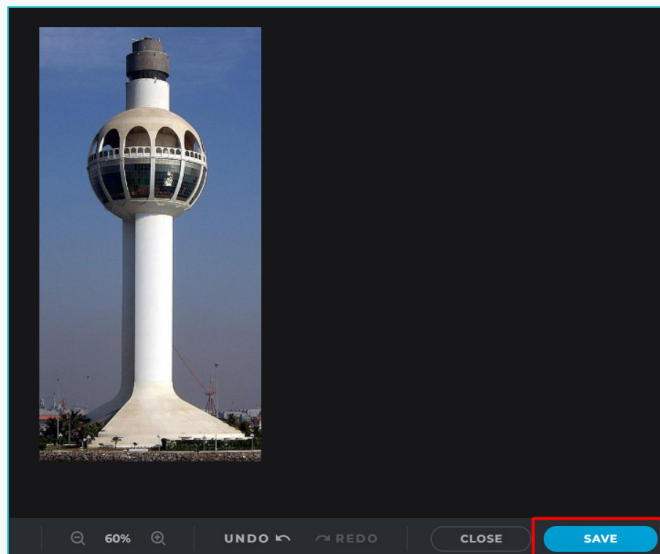
- 11 Click on Apply to view the final image.



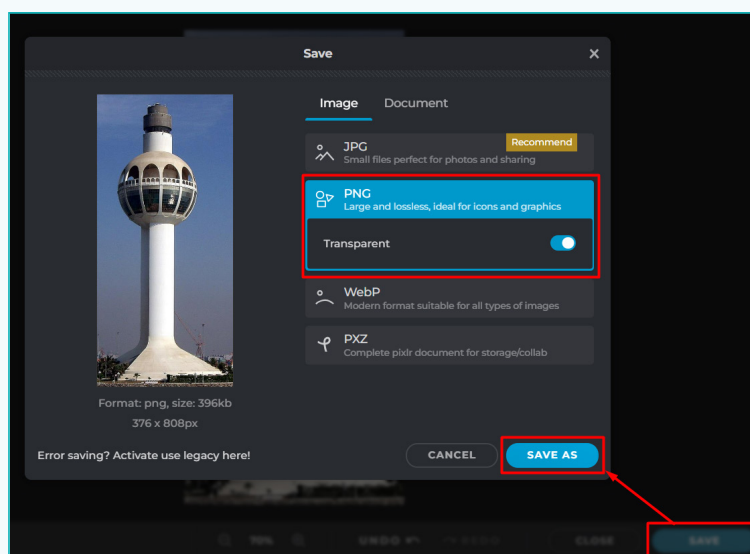
Activity

To save the image

- 1 Click on the Save button in the **Actions pane**. The Actions Pane has commands to undo or redo an action, as well as to save or close the image.



- 2 When the Save button is clicked, it opens a window to set the parameters of the image. This lets us set the name of the image file, file type, and its quality.

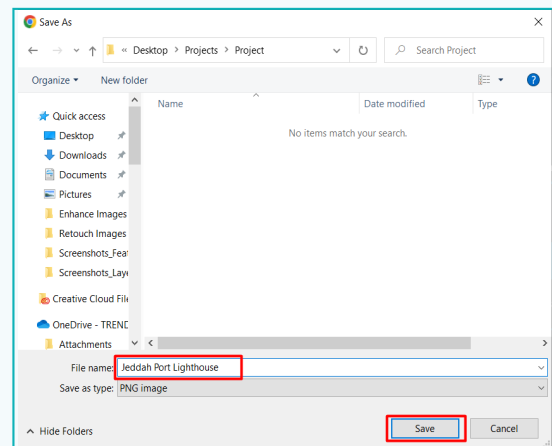
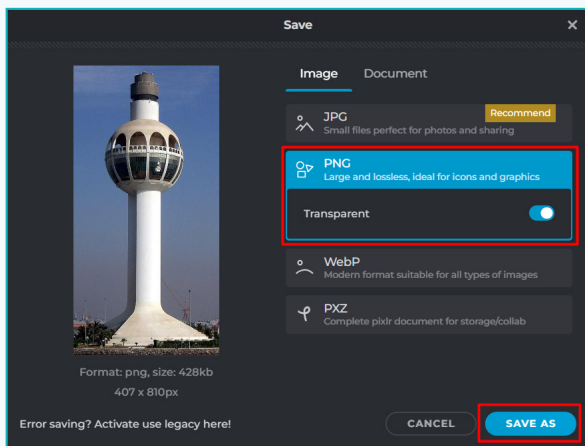


Activity

Observe the different file types available to save the image in

- **JPEG (.jpg, .jpeg):** JPEG stands for Joint Photographic Experts Group. JPEG is a popular format for digital cameras. This makes it ideal for use on the internet. When we save an image as a .jpg file, there is a slight loss in quality. It is not too clear when we view it on a screen.
- **PNG (.png):** PNG stands for Portable Network Graphics. PNG files can handle up to 16 million colours. There is no quality loss when we save an image in .png format.
- **PXD:** Pixlr Editor has its own file format called **PXD**, which preserves image edits. They can be modified later.

3 For this activity, we will save our image as a PNG file.



Bytes

GIF(.gif): GIF stands for Graphics Interchange Format. They are widely used to create animation because they do not lose their quality after enhancement.

Activity

Final output



Our image is ready to use! Set it as a wallpaper on a mobile phone.



Exercise

Fill in the blanks

- 3 Image processing is a method to perform some operations on an image, to _____ or to _____ some useful information from it.
- 4 PNG stands for _____.
- 5 JPEG stands for _____.

State whether True or False

- 6 The total count of pixels in an image is called the dimension of the image.
▲ _____
- 7 Tool Bar consists of all the tools that the application has to offer to edit an image.
▲ _____
- 8 The Actions Pane has commands to undo or redo an action, as well as to save or close the image.
▲ _____
- 9 There is a slight loss in quality when we save an image in .png format.
▲ _____

In this lesson, we saw the different types of images. We learnt that digital images are made of units called Pixels and can be enhanced with different tools. We also learnt how to edit a landscape image to make it into a vertical wallpaper.

8. Features of Images

This was easy! Now I know how to change the size of an image.

Show me other pictures that you have captured.



Tech Flash

- **Image processing** : Image processing is a process that analyses and manipulates images.
- **Cropping** : Cropping removes unwanted outer areas from a photographic or illustrated image.
- **PXD** : Pixlr Editor has its own file format called PXD, which preserves image layers for future editing.
- **Pixel** : A Pixel (short for picture element) is a single point in a digital picture. Images made up of pixels are also called rasterized images.